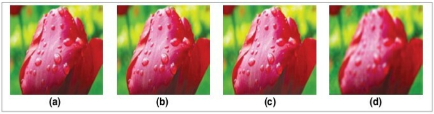

Fig. 3: (a) Original image and (b)-(d) compressed versions using DCT

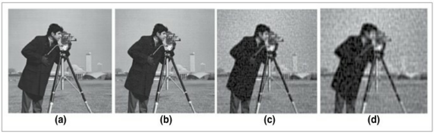

Fig. 4: (a) Original image and (b)-(d) compressed images

transform (DCT).

The concept of sparse domain is illustrated in Fig. 2 in a simple way. Fig. 2(a) shows a high-definition original image that occupies 2.3MB of space.

In spatial domain, this image is represented as a matrix of numbers, which are basically image-intensity levels. The plot of intensity levels, known as histogram of the image, is shown in Fig. 2(b). It can be observed from Fig. 2(b) that these intensity levels vary across a large range from 0 to 255. However, if you transform the same image using wavelet, curvelet, DCT or SVD domain, you get the plot of respective coefficients as shown in Fig. 2(c).

It can be observed that the same image can be represented using fewer coefficients as most of the coefficients in these domains are nearly zero. Hence, discarding these nearly-zero coefficients and retaining only non-zero coefficients reduces the memory space required to store these coefficients, which, in turn, helps in compressing the image.

Here, we use DCT for image compression. Please note, it is not known what algorithm WhatsApp software is utilising for compressing its images and videos. Our intention in this article is to present the underlying concept behind the operation discussed in Fig. 1. Figure 3(a) shows a flower image that occupies 2.3MB of storage space. Using the DCT-based image compression algorithm, we obtained compressed images of sizes 392kB, 274kB and 223kB as shown in Figs 3(b)-(d), respectively. It can be seen, as the size of an image is compressed, artifacts tend to occur near the edges of the image. This is clearly visible in Fig. 3(d) where significant artifacts are visible.

The compression ratio is defined as:

K = Uncompressed size of an image/Compressed size of an image

For the images in Figs. 3(b)-(d), the value of K is obtained as 5, 8 and 10, respectively.

It can be observed from Fig. 3 that the images obtained after compression occupied less space and yet were good enough for visual inspection. Similar analysis is performed on grayscale image of a cameraman and the results are shown in Fig. 4.

Here, the compression ratios for the images shown in Figs 4(b)-(d) are given as 5, 6 and 7, respectively. It can be seen, as the compression ratio increases from left to right, the blocking artifacts tend to appear in an image. This can be clearly seen from Fig. 4(d).

Testing procedure

The program (code.m) can be used for colour and grayscale images both. Tulip.jpeg colour image and cameraman.jpeg grayscale images were used during the testing of this program. You need to select one image (either colour or grayscale) at a time.

1. Install MATLAB R2013a or later version in your system. Open the code.m file
2. If colour image is to be compressed, line number 11 of the code.m file has to be uncommented. It is already uncommented for the given program
3. If grayscale image is to be compressed, line number 14 of the code.m file has to be uncommented.
4. Once you have selected either step 2 or step 3, select Run command button. Then the program prompts you to enter the threshold value. After entering this value followed by pressing Enter key, you need to wait for some time till the compressed image pops up on the screen.
5. For a coloured image, choose any one threshold value (e.g., 5, 50 or 500) for generating images shown in Figs 3(b)-(d), respectively.
6. For grayscale images, choose one threshold value (e.g., 10, 60 or 100) for generating images shown in Figs 4(b)-(d), respectively. EFY

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*


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